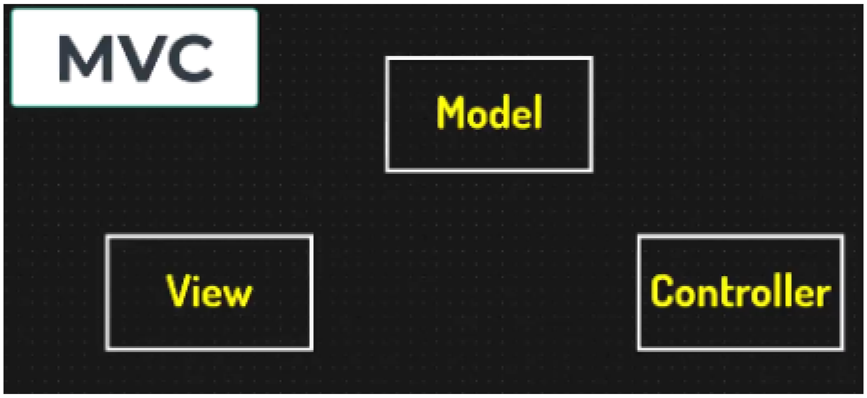
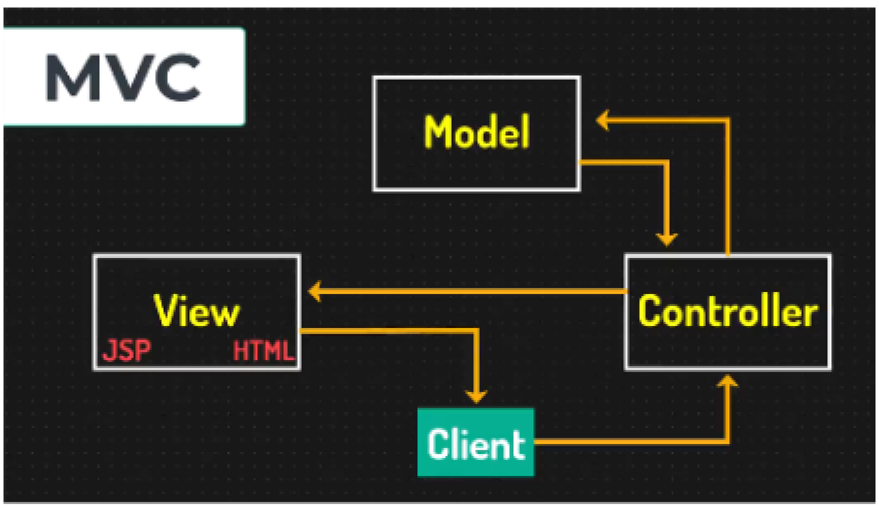
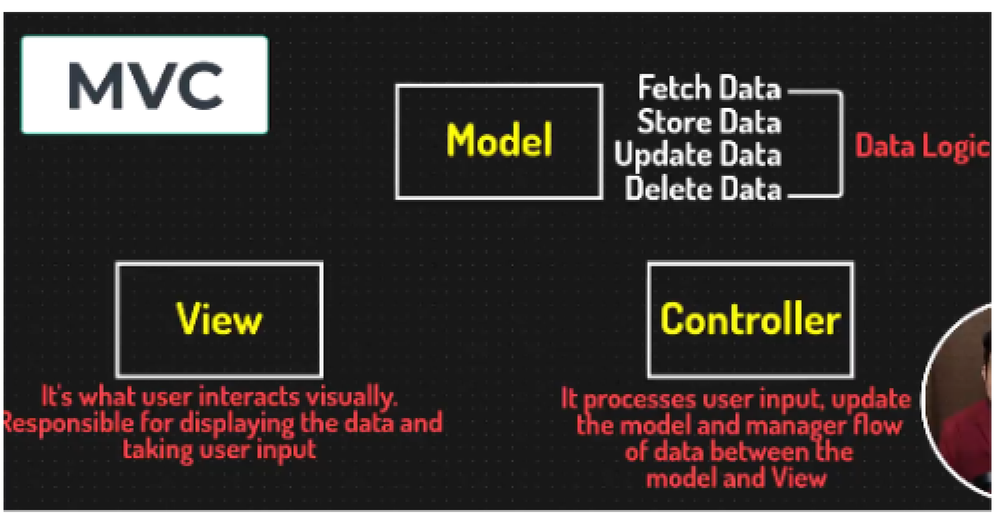
Introduction to MVC







👉MVC Overview:

- **Purpose:** The MVC pattern is used to separate an application into three interconnected components, helping to manage complex applications by dividing responsibilities and promoting organized code.

👉Components:

1. **Model:**
    - **Role:** Represents the data and business logic of the application.
    - **Responsibilities:**
        - Manages data retrieval, storage, and manipulation.
        - Notifies the view of any changes in the data.
        - Encapsulates the business rules and logic.
2. **View:**
    - **Role:** Responsible for presenting the data to the user and rendering the user interface.
    - **Responsibilities:**
        - Displays data received from the model.
        - Provides a way for users to interact with the application.
        - Updates the user interface in response to model changes.
3. **Controller:**
    - **Role:** Acts as an intermediary between the model and view, handling user input and updating the model.
    - **Responsibilities:**
        - Receives user input from the view.

- Processes input, performs actions (e.g., updating data), and interacts with the model.
- Updates the view based on changes to the model or user actions.

👉**Workflow:**

1. **User Interaction:** The user interacts with the view (e.g., submits a form).
2. Controller: The controller receives the input from the view, processes it, and interacts with the model.
3. **Model Update:** The model updates its state based on the controller's actions and may notify the view of changes.
4. **View Update:** The view retrieves the updated data from the model and refreshes the display for the user.

👉**Benefits of MVC:**

- **Separation of Concerns:** Each component has a distinct responsibility, making the system easier to manage, test, and scale.
- **Modularity:** Changes in one component (e.g., changing the view) have minimal impact on others (e.g., model or controller).
- **Maintainability:** Improved maintainability due to clear separation of business logic, data, and user interface.
- **Reusability:** Components can be reused across different parts of the application or in different projects.

👉**Summary:**

- Model handles data and business logic.

- View manages user interface and presentation.
- Controller processes user input and coordinates between the model and view.